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PCB Board Defect Detection Algorithm Based on Improved YOLOv8

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***Abstract*—To address the issues of insufficient fine-grained defect feature extraction and limited detection accuracy in traditional YOLOv8 algorithm models for PCB industrial board defect detection, this paper proposes a PCB board defect detection method based on an improved YOLOv8 network. This method introduces partial convolutional modules and multi-scale channel attention mechanisms into the head network of the original YOLOv8 algorithm framework to enhance the model's feature extraction capabilities and multi-scale information fusion capabilities for defect regions. The PConv module effectively improves the model's perception of local features in defect regions, while the MLCA mechanism enhances the model's ability to focus on features across different scale channels, thereby improving detection accuracy and robustness. Experiments show that the improved algorithm model achieves an average accuracy of 87.3% and a recall rate of 82.1%, which are 4.1% and 2.9% higher than the original YOLOv8 model, respectively. This study provides an efficient and high-precision solution for PCB board defect detection, which is of great value for practical industrial detection.**

Keywords—component, YOLOv8, PCB, Image processing, Object detection

I. INTRODUCTION

The rapid development of industry contributes to national economic prosperity, and PCB industrial boards serve as the foundational components for many critical electronic devices. Traditional PCB industrial board inspection has typically relied on manual inspection or simple machine screening. However, manual inspection suffers from low efficiency, susceptibility to fatigue, and strong subjectivity, while machine screening struggles to accurately identify complex microscopic defects, particularly in high-density, multi-layer circuit boards[1]. In contrast, an automated inspection system built using deep learning algorithms can achieve high-precision identification and classification of defect features in PCB images, thereby enhancing the objectivity and consistency of defect detection. Additionally, through the training and analysis of large-scale defect image data, it is possible to achieve visual tracking and statistical analysis of defect types, helping enterprises achieve closed-loop quality management in production processes[2]. This study aims to further improve the quality control level of the electronics manufacturing industry and promote the transformation and upgrading of smart manufacturing[3].

With the continuous advancement of image processing and deep learning technologies, intelligent detection has found increasingly widespread application in industrial fields, resulting in significant improvements in recognition accuracy and processing efficiency. However, due to factors such as the complexity of PCB board patterns, the diversity of defect shapes, and the blurred boundaries between defects, current automated detection methods still face numerous challenges in terms of robustness and small-sample detection, necessitating further optimization and breakthroughs. The YOLO series of object detection algorithms, as one of the core methods for achieving object recognition in various scenarios through deep learning, has garnered widespread attention and application due to its advantages of lightweight structure and high detection accuracy. Currently, an increasing number of researchers are attempting to integrate such machine learning algorithms into the field of behavior recognition and detection to enhance the system's intelligence level and detection efficiency. Currently, an increasing number of researchers are attempting to incorporate this class of machine learning algorithms into the field of behavior recognition and detection to enhance the system's intelligence level and detection efficiency. Yu et al. [4] enhanced the YOLOv8 algorithm and established the YOLO-Bo network model. This algorithm introduces BotNet into the feature extraction network, combining convolutions and self-attention, with experimental results showing good accuracy. Gu et al. [5] use the BiFPN in the feature fusion stage to achieve multi-scale feature extraction and fusion, while adding an RFB module before each of the three decoupled heads to expand the receptive field and improve the detection accuracy of small defects in PCBs. Zhang et al. [6] adopted BiFPN as their feature fusion architecture, utilizing bidirectional multi-scale feature fusion to enable the model to more effectively distinguish between defects with similar attributes and those exhibiting significant scale differences. Experimental results demonstrated that this method significantly improved target recall rates. Ren et al. [7] proposed a PCB defect detection method based on an improved YOLOv8 model. The backbone network employs the lightweight model PP-LCNet, whose depth-separable convolutions and other designs reduce the number of

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parameters and computational complexity, adapting to the characteristics of circuit board images and efficiently extracting key features. Lan et al. [8] added a cross-scale fusion module between the backbone network and the neck network. The CFM module is used for the fusion of features from adjacent three layers, Experimental results show that the bestmAP@0.5of the improved YOLOv8n algorithm reaches 96.6%. Li et al. [9] proposed a C2F-DC module based on dynamic convolution to replace the C2f module in the backbone network, enhancing the model's feature extraction capability by adaptively generating convolution kernel weights. Additionally, they introduced an HSD module to preserve defect details during downsampling. Experimental results show that the average accuracy improved by 2.4%, and the recall rate improved by 3.1%. Yan et al. [10] proposed the SiLU-SE attention module to enhance feature maps, improving detection accuracy at a lower computational cost. Experimental results show that compared to the original YOLOv8n, the improved model size decreased by approximately 77.01%, and the number of model parameters decreased by approximately 84%.

The deep learning algorithms proposed in the aforementioned literature often suffer from insufficient detection accuracy when dealing with scenarios involving a high density of targets. Additionally, current mainstream object detection algorithms generally feature complex structures and a large number of model parameters, resulting in high dependence on computational resources, which limits their deployment and application in edge devices or resource-constrained environments. To address this issue, this paper proposes an object detection algorithm that balances accuracy and efficiency. This algorithm incorporates the MLCA attention mechanism to enhance the network's ability to perceive and extract key target features. It also integrates the PConv module to effectively reduce computational overhead while improving the model's ability to perceive local features, thereby meeting the requirements of practical applications.

II. Method

A. YOLOv8 Network

In early 2023, Ultralytics launched YOLOv8, a major upgrade to the YOLO series that marks another breakthrough in object detection technology. This version not only continues the excellent performance of previous YOLO models but also incorporates the latest research findings in deep learning and computer vision. As a single-stage object detection algorithm, YOLOv8 offers efficient and fast object localization and recognition capabilities. Its network structure primarily consists of three major modules: the Backbone (main network), the Neck (neck network), and the Head (head network). The Backbone is responsible for extracting deep-level key features from the input image; the Neck, located between the Backbone and the Head, handles multi-scale feature fusion and information enhancement; and the Head serves as the model's output module, generating the final detection results to complete the object localization and classification tasks. In this study, the MLCA attention mechanism and PConv module were incorporated into the Neck network to obtain the improved YOLOv8 network. The structure of the improved YOLOv8 network is shown in Figure 1.

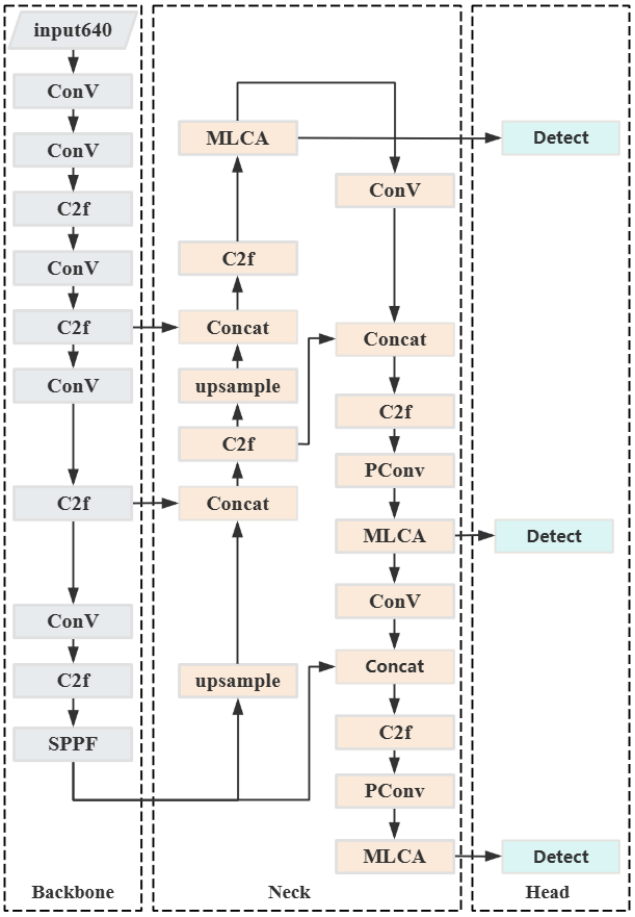


Fig. 1. Improved YOLOv8 Network Structure

B. YOLOv8 Improvement Methods

1) MLCA Attention Mechanism

The MLCA attention mechanism achieves multi-scale enhancement of feature maps by jointly modeling local and global context information. First, the input feature maps are subjected to local average pooling (LAP) and global average pooling (GAP) to capture spatial information at different levels. Subsequently, the features from both branches undergo one-dimensional convolution for channel compression and feature transformation, thereby enhancing expressive power while maintaining the spatial dimension. The transformed features are further rearranged to adapt to subsequent fusion operations. In the local path, the transformed local pooling features are interacted with the original input features through element-wise multiplication, achieving selective enhancement of key spatial regions. In the global path, the processed global features are fused with the local enhancement results through element-wise addition, further injecting global contextual information to enhance the model's semantic perception capabilities. Finally, the fused feature maps are restored to their original spatial resolution through unpooling operations, completing the reconstruction of multi-level context-aware features. The MLCA attention mechanism effectively enhances the model's ability to express multi-scale semantic and structural features through its joint local and global modeling approach. The structure of the MLCA attention mechanism is shown in Figure 2.

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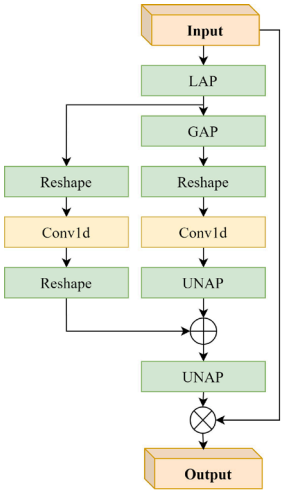


Fig. 2. MLCA Attention Mechanism Structure

2) PConv Module

The PConv module (Partial Convolution Module) is an efficient feature transformation structure designed to enhance the model's ability to express key features while maintaining its lightweight nature. Unlike traditional full-channel convolution, PConv employs a strategy where only a portion of the input feature map's channels undergo convolution, while the remaining channels remain unchanged or participate in subsequent fusion via skip connections. This significantly reduces computational cost and enhances flexibility in feature selection. The principle diagram of the PConv module is shown in Figure 3, and its computational method is described in Equations 1-3.

$$X = [X_a, X_b] \tag{1}$$

$$Y_a = Conv(X_a) \tag{2}$$

$$Y = [Y_a, X_b] \tag{3}$$

First, divide the feature map into channel dimensions, perform convolution operations on a portion of it, and then concatenate the final results for output.

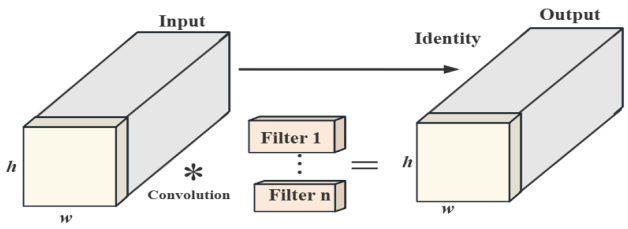


Fig. 3. PConv Module Schematic Diagram

III. Results & Discussion

A. Preparation for the Experiment

This experiment uses the deep learning Pytorch framework and is trained on a Linux server. The graphics card model used is RTX4060, with 8G of graphics memory and CUDA version 12.6. The specific model training parameters are shown in Table I.

TABLE I. Model Training Parameter Table

Model parameters	Configuration
imgsz	640px
momentum	0.9
optimizer	SGD
Epoch	300
batch size	16
lr	0.001

B. Dataset Introduction

This experiment uses a private PCB industrial board defect detection dataset, which contains a total of 1,563 images. The dataset is divided into training, validation, and test sets in a 7:2:1 ratio. There are six target detection categories in this dataset, namely: “missing hole”, “mouse bite”, “open circuit”, “short”, “spurious copper”, and “spur”.

C. Evaluation Indicators

The performance evaluation of the model in this experiment is mainly based on precision P, recall rate R, and mean average precision mAP. The formulas for calculating each indicator are as follows:

$$R = TP / (TP + FN) \tag{4}$$

$$P = TP / (TP + FP) \tag{5}$$

$$mAP = \frac{\sum AP}{N} \tag{6}$$

Precision (P) measures the proportion of actual positives among the samples predicted as positive by the model; recall (R) indicates the proportion of all true positive samples successfully identified by the model. mAP (mean average precision) is the average precision of all categories and is an important indicator for evaluating the overall performance of object detection models.

D. Analysis of Experimental Results

This experiment proposes an improved YOLOv8 object detection algorithm, introducing the MLCA attention mechanism into the backbone network and adding the original Neck part of YOLOv8 to the PConv module. After iterating the model 300 times, the results are shown in Figure 4. The model's performance stabilizes after 150 iterations, with an average accuracy of 87.3%.

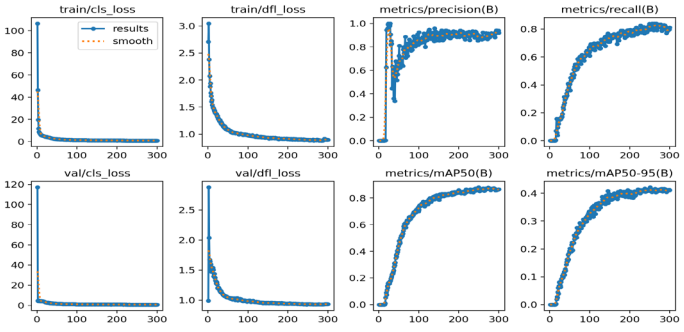


Fig. 4. Improved Algorithm Model Performance

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E. Comparative Experiments

This experiment compares the proposed improved algorithm with the original YOLOv8 model, primarily evaluating performance in terms of detection accuracy and model parameter count. The comparison results are shown in Table II. The improved model achieves a significant increase in detection accuracy compared to the original model while maintaining the same number of model parameters. This indicates that it offers higher efficiency and accuracy in practical applications. Additionally, to further validate the effectiveness of each improved module, ablation experiments were designed and conducted. The ablation experiments were set up with four comparison schemes. Experiment 1 used the original YOLOv8 model, Experiment 4 used the fully improved model incorporating the PConv module and MLCA attention mechanism, and the remaining two groups were intermediate schemes introducing a single module each. The experimental results are shown in Table III, where "√" indicates that the corresponding module was introduced. The results indicate that both the PConv and MLCA mechanisms positively contribute to improving model performance to varying degrees, further validating the rationality and effectiveness of the proposed structural design.

TABLE II. COMPARATIVE EXPERIMENT

Name	*P%*	*R%*	*mAP%*	*GFLOPs*	*Parameters*
YOLOv8	82.1	79.2	83.2	8.2	3.01M
Model	**87.4**	**82.1**	**87.3**	**8.3**	**3.11M**

TABLE III. ABLATION EXPERIMENT

Experiment	MLCA	PConv	*P%*	*R%*	*mAP%*
1			82.1	79.2	83.2
2		√	85.6	81.1	84.9
3	√		86.3	81.9	86.1
4	√	√	**87.4**	**82.1**	**87.3**

F. Visualization of Results

In this experiment, the trained model was used to perform object detection on the PCB defect dataset, and the detection results are shown in Figure 5. As can be seen from Figure 5, the improved model performs well in identifying various types of PCB defects, accurately locating and classifying defect areas, thereby validating its effectiveness in practical applications.

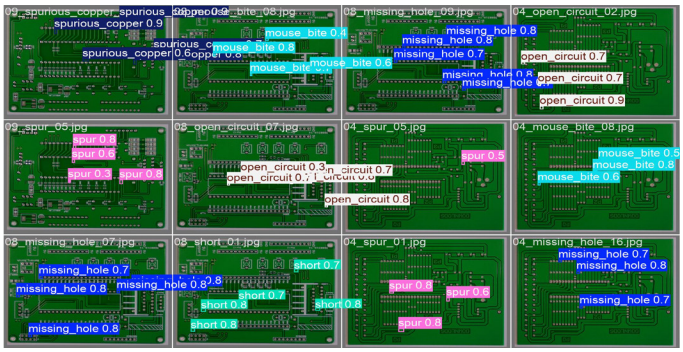


Fig. 5. Visualization display

IV. CONCLUSIONS

This paper addresses the limitations of the traditional YOLOv8 model in PCB defect detection tasks, such as insufficient defect feature extraction and limited performance in identifying dense targets, by proposing an improved YOLOv8 object detection algorithm. The improved model incorporates a PConv convolution structure to enhance feature extraction robustness and sensitivity to local information. Additionally, an MLCA attention mechanism is introduced to strengthen the model's ability to model multi-scale features, thereby improving detection performance for small defects. Experimental results demonstrate that the proposed method not only enhances detection accuracy but also exhibits excellent generalization capabilities and practical application value. Future work will continue to optimize the model structure to ensure more stable and efficient performance across diverse industrial scenarios.

REFERENCES

[1] X. Huang and W. Li, "A Novel PCB Defect Detection Network Based on the Improved YOLOv8 with Fusion of Hybrid Attention Transformer and Bidirectional Feature Pyramid Network," 2024 4th International Conference on Artificial Intelligence, Robotics, and Communication (ICAIRC), Xiamen, China, 2024, pp. 207-211.

[2] Y. Ai and T. Ye, "Surface Defect Detection Algorithm for PCB Based on Improved YOLOv8," 2024 8th International Conference on Electrical, Mechanical and Computer Engineering (ICEMCE), Xi'an, China, 2024, pp. 1421-1425.

[3] A. Anish, S. R, A. H. Malini and T. Archana, "Enhancing Surveillance Systems with YOLO Algorithm for Real-Time Object Detection and Tracking," 2023 2nd International Conference on Automation, Computing and Renewable Systems (ICACRS), Pudukkottai, India, 2023, pp. 1254-1257.

[4] J. Yu and Y. bin Gao, "Defect Detection on PCB Bare Boards Based on Improved YOLOv8 Algorithme," 2024 3rd International Conference on Cloud Computing, Big Data Application and Software Engineering (CBASE), Hangzhou, China, 2024, pp. 526-529.

[5] X. Gu, C. Cao and Z. Xu, "Research on PCB Defect Detection Algorithm Based on Improved YOLOv8," 2024 6th International Conference on Frontier Technologies of Information and Computer (ICFTIC), Qingdao, China, 2024, pp. 1103-1108.

[6] R. Zhang, S. Kai, L. Yin and C. Pan, "Research on PCB Defect Detection Algorithm Based on Improved YOLOv8," 2024 5th International Conference on Artificial Intelligence and Computer Engineering (ICAICE), Wuhu, China, 2024, pp. 967-971.

[7] H. Ren, Y. Wang, X. Song and S. Wang, "PPW-YOLO: An Improved PCB Defect Detection Model Based on yolo," 2025 5th International Conference on Sensors and Information Technology, Nanjing, China, 2025, pp. 897-900.

[8] H. Lan, H. Zhu, R. Luo, Q. Ren and C. Chen, "PCB Defect Detection Algorithm of Improved YOLOv8," 2023 8th International Conference on Image, Vision and Computing (ICIVC), Dalian, China, 2023, pp. 178-183.

[9] Y. Li and W. Yang, "YOLO-DHC: A Multi-scale method for PCB defect detection," 2024 4th International Symposium on Artificial Intelligence and Intelligent Manufacturing (AIIM), Chengdu, China, 2024, pp. 314-319.

[10] M. Yan, Y. Fan, Y. Jiang and Z. Fang, "A Fast Detection Algorithm for Surface Defects of Bare PCB Based on YOLOv8," 2023 3rd International Conference on Electronic Information Engineering and Computer Communication (EIECC), Wuhan, China, 2023, pp. 559-564.

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